

Final Portfolio Project

Seth Bonney and Cuong Tran

Research Question

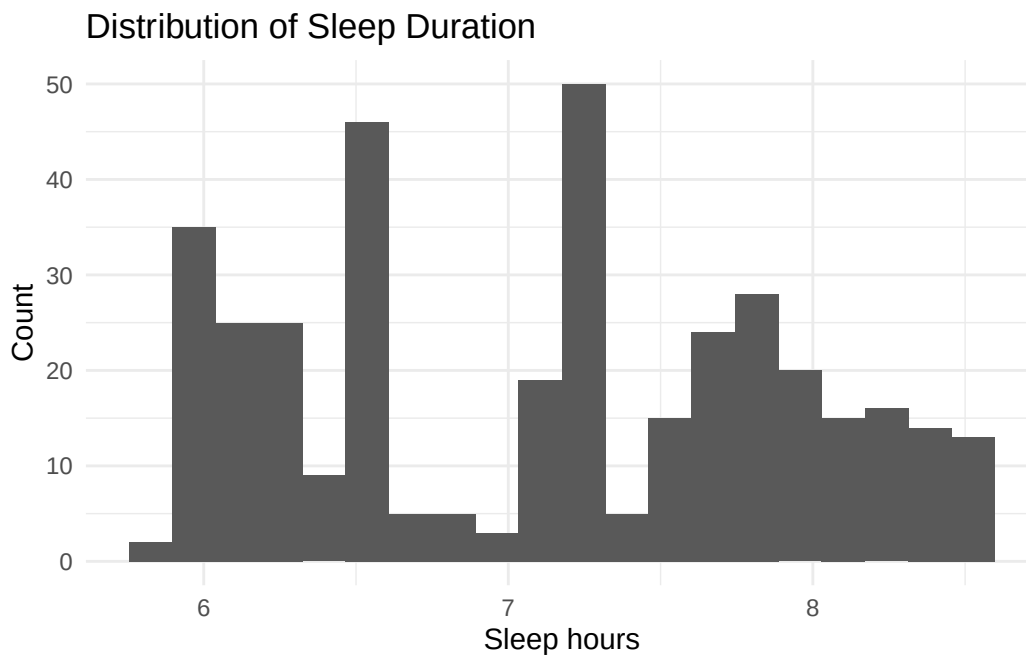
Will come up soon

```
full_data_clean <- full_data |>
  select(-c(sleep_hours)) |>
  relocate(sleep_duration)
```

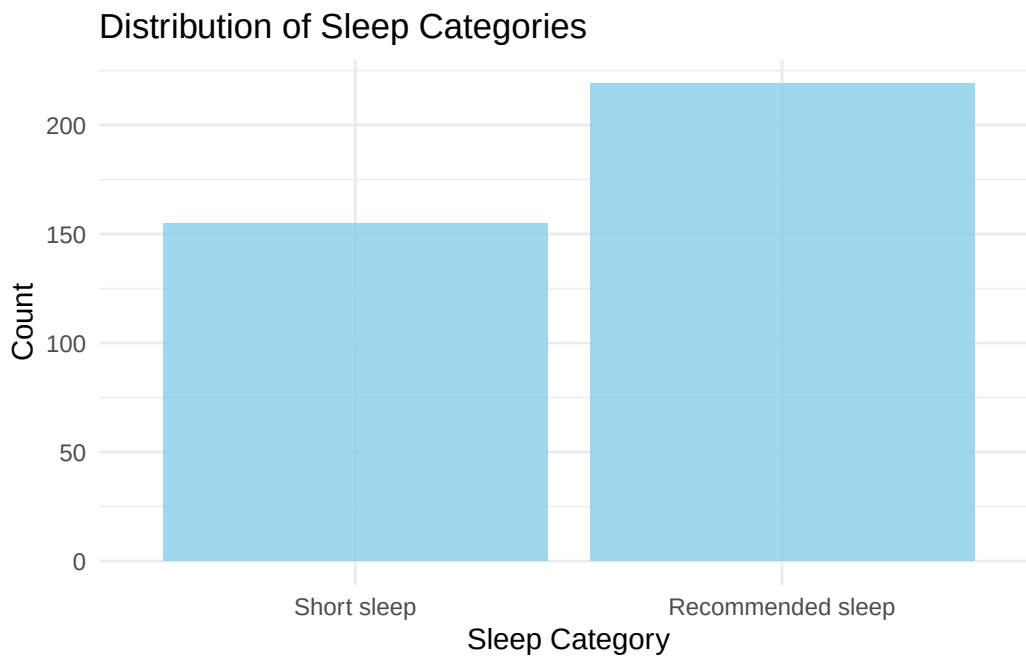
Research question 1

#These are graphs from the sleep dataset

```
library(ggplot2)
ggplot(sleep, aes(sleep_duration)) +
  geom_histogram(bins = 20) +
  theme_minimal() +
  labs(
    title = "Distribution of Sleep Duration",
    x = "Sleep hours",
    y = "Count"
  )
```



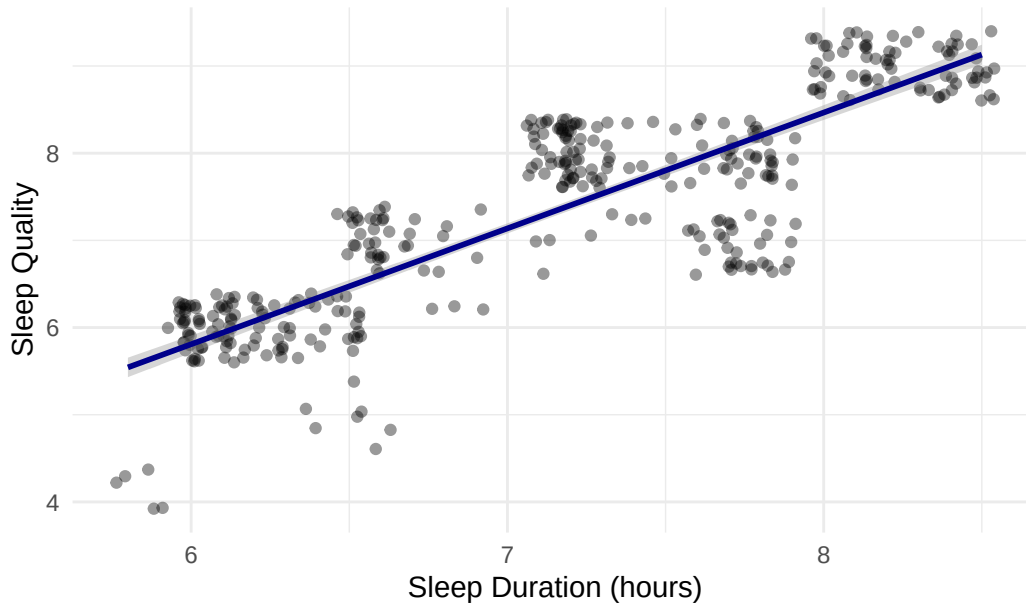
```
ggplot(sleep, aes(sleep_category)) +
  geom_bar(fill = "skyblue", alpha = 0.8) +
  theme_minimal() +
  labs(
    title = "Distribution of Sleep Categories",
    x = "Sleep Category",
    y = "Count"
  )
```



```
ggplot(sleep, aes(x = sleep_duration, y = sleep_quality)) +
  geom_jitter(alpha = 0.4) +
  geom_smooth(method = "lm", se = TRUE, color = "darkblue") +
  theme_minimal() +
  labs(
    title = "Sleep Duration vs Sleep Quality",
    x = "Sleep Duration (hours)",
    y = "Sleep Quality"
  )
```

`geom\_smooth()` using formula = 'y ~ x'

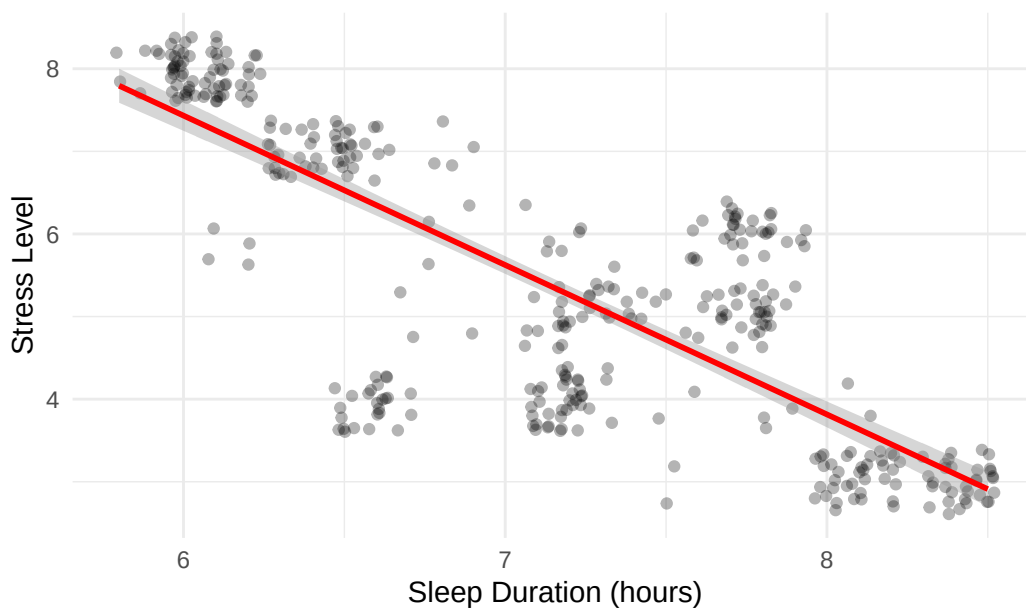
Sleep Duration vs Sleep Quality



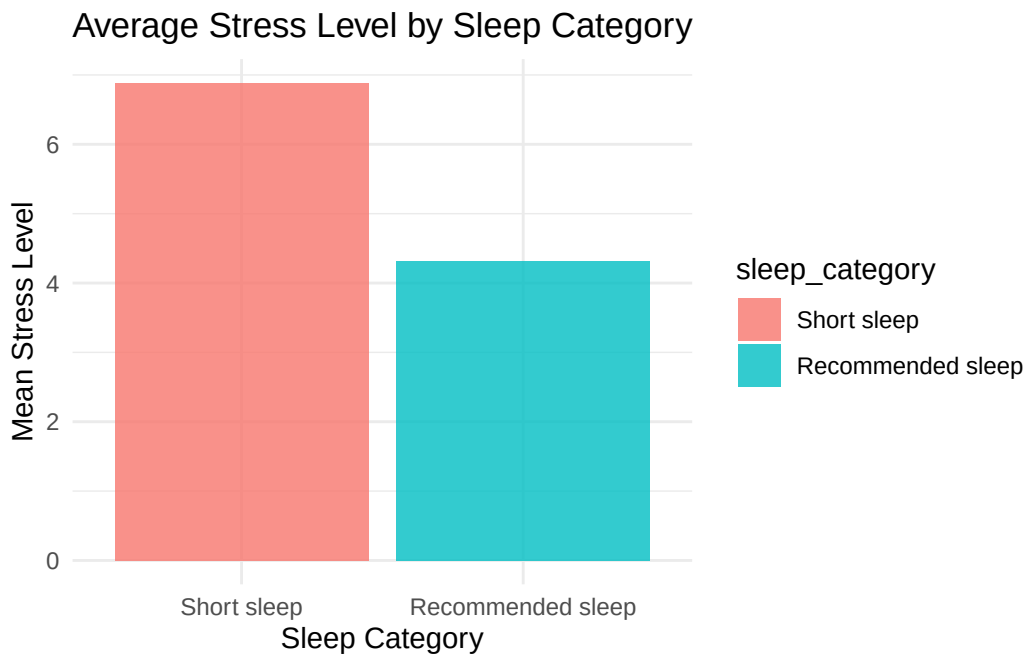
```
ggplot(sleep, aes(x = sleep_duration, y = stress_level)) +
  geom_jitter(alpha = 0.3) +
  geom_smooth(method = "lm", se = TRUE, color = "red") +
  theme_minimal() +
  labs(
    title = "Sleep Duration vs Stress Level",
    x = "Sleep Duration (hours)",
    y = "Stress Level"
  )
```

`geom\_smooth()` using formula = 'y ~ x'

Sleep Duration vs Stress Level



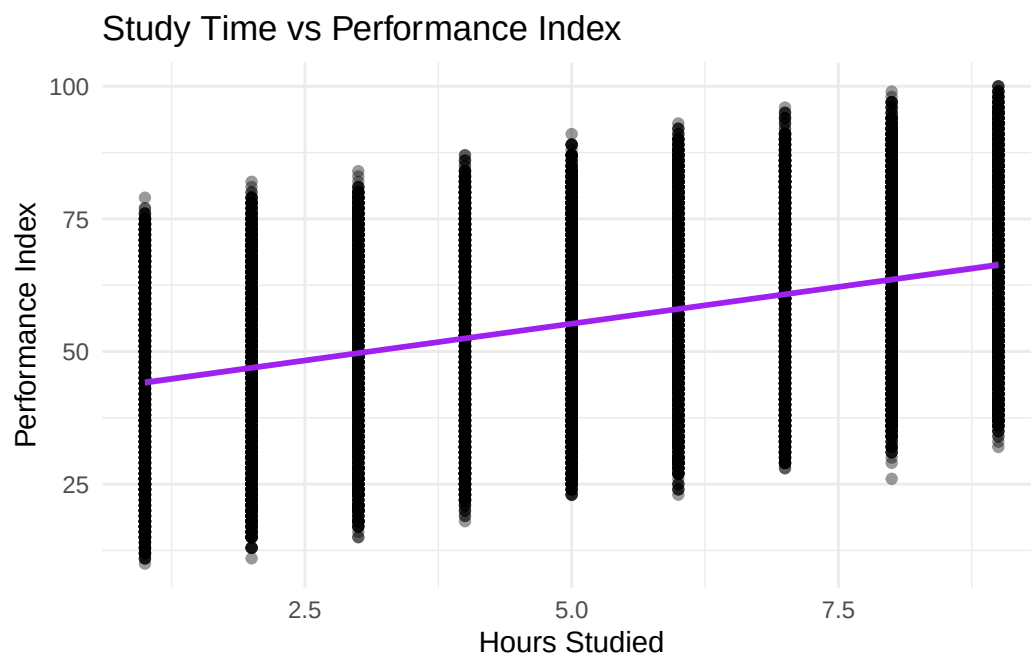
```
sleep |>
  group_by(sleep_category) |>
  summarise(mean_stress = mean(stress_level)) |>
  ggplot(aes(sleep_category, mean_stress, fill = sleep_category)) +
  geom_col(alpha = 0.8) +
  theme_minimal() +
  labs(
    title = "Average Stress Level by Sleep Category",
    x = "Sleep Category",
    y = "Mean Stress Level"
  )
```



graphs for the student performance dataset

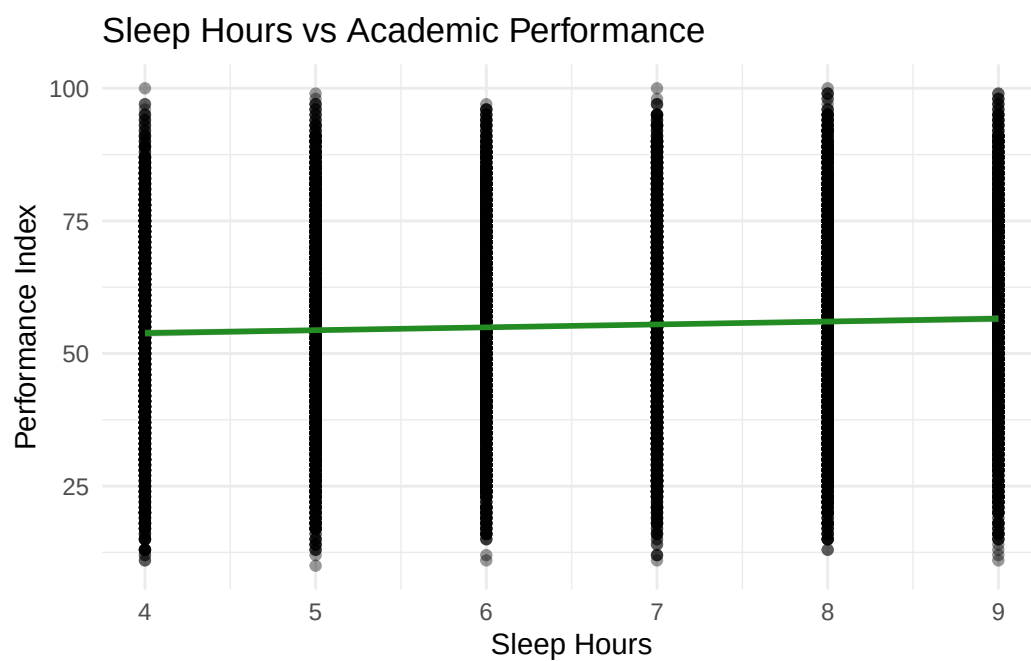
```
ggplot(student, aes(x = hours_studied, y = performance_index)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = TRUE, color = "purple") +
  theme_minimal() +
  labs(
    title = "Study Time vs Performance Index",
    x = "Hours Studied",
    y = "Performance Index"
  )
```

`geom\_smooth()` using formula = 'y ~ x'



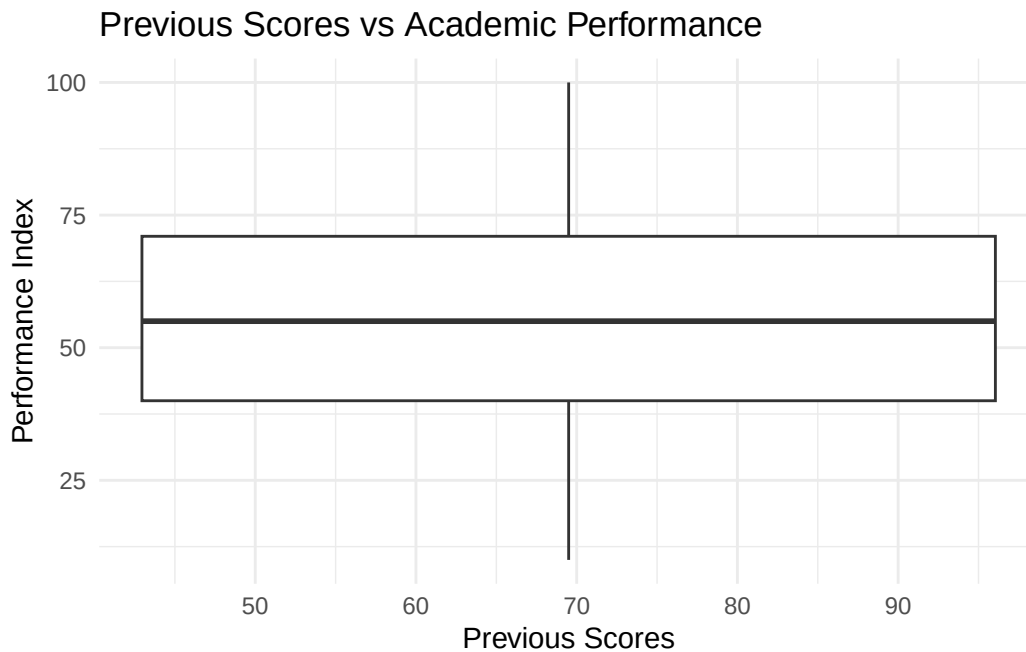
```
ggplot(student, aes(x = sleep_hours, y = performance_index)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = TRUE, color = "forestgreen") +
  theme_minimal() +
  labs(
    title = "Sleep Hours vs Academic Performance",
    x = "Sleep Hours",
    y = "Performance Index"
  )
```

`geom\_smooth()` using formula = 'y ~ x'



```
ggplot(student, aes(x = previous_scores, y = performance_index)) +
  geom_boxplot() +
  theme_minimal() +
  labs(
    title = "Previous Scores vs Academic Performance",
    x = "Previous Scores",
    y = "Performance Index"
  )
```

Warning: Continuous x aesthetic
i did you forget `aes(group = ...)`?

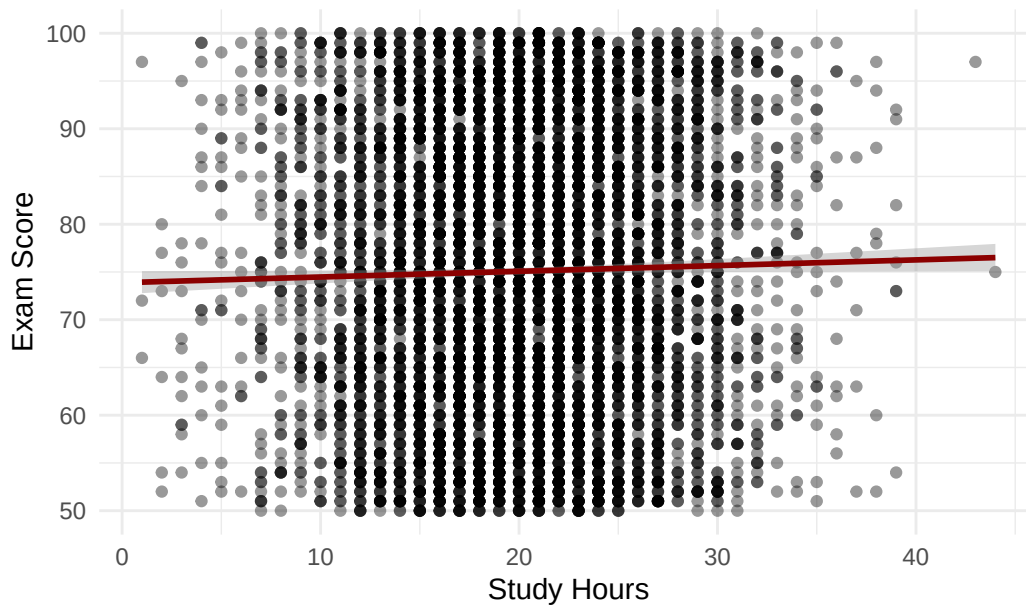


Graphs for the factor dataset

```
ggplot(factors, aes(x = study_hours, y = score)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = TRUE, color = "darkred") +
  theme_minimal() +
  labs(
    title = "Study Hours vs Exam Score",
    x = "Study Hours",
    y = "Exam Score"
  )
```

`geom\_smooth()` using formula = 'y ~ x'

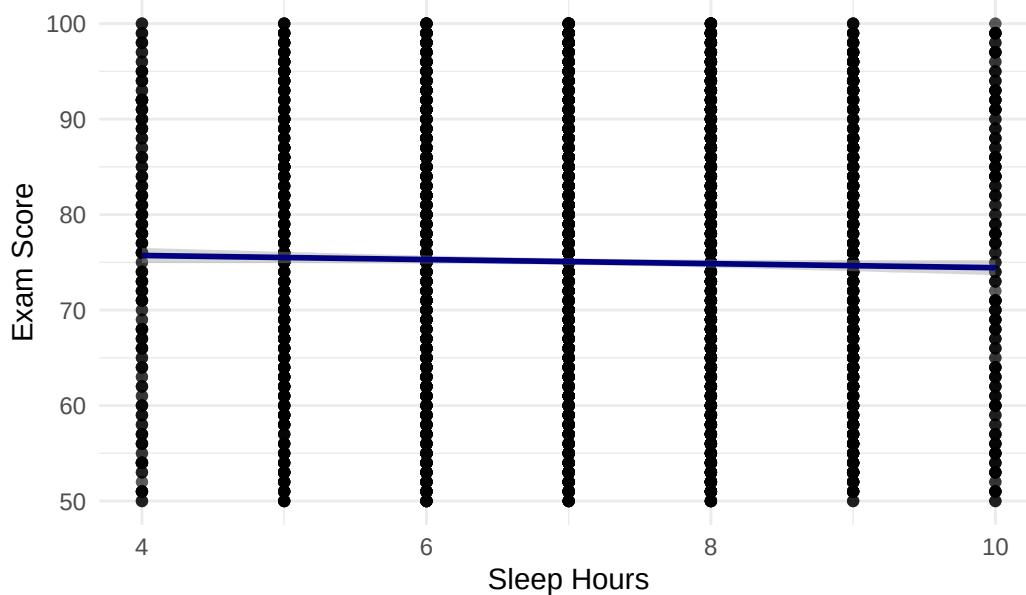
Study Hours vs Exam Score



```
ggplot(factors, aes(x = sleep_hours, y = score)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = TRUE, color = "navy") +
  theme_minimal() +
  labs(
    title = "Sleep Hours vs Exam Score",
    x = "Sleep Hours",
    y = "Exam Score"
  )
```

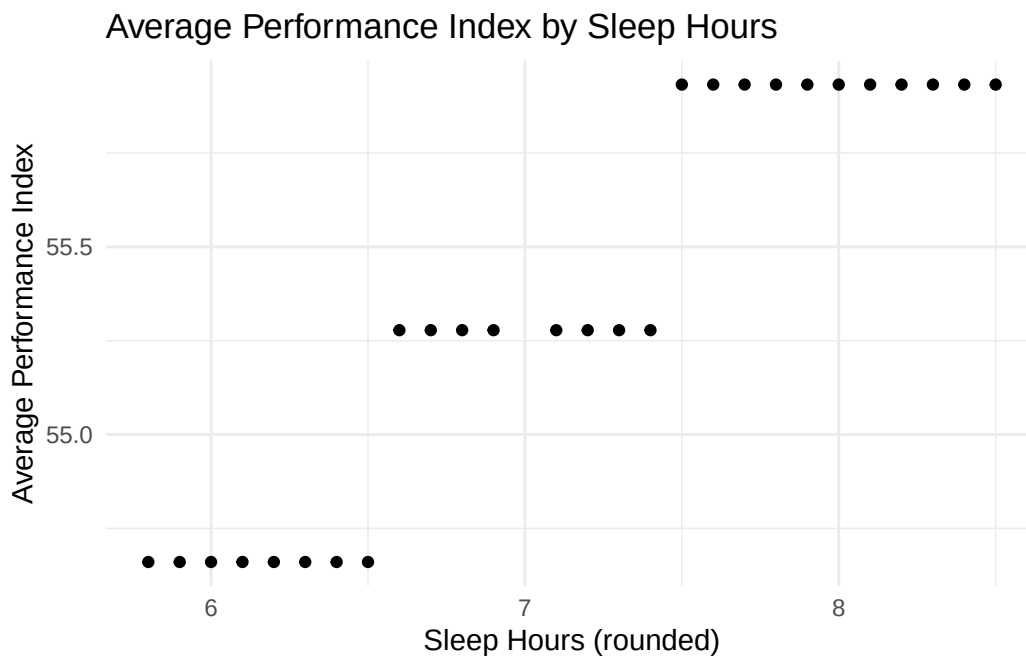
`geom\_smooth()` using formula = 'y ~ x'

Sleep Hours vs Exam Score

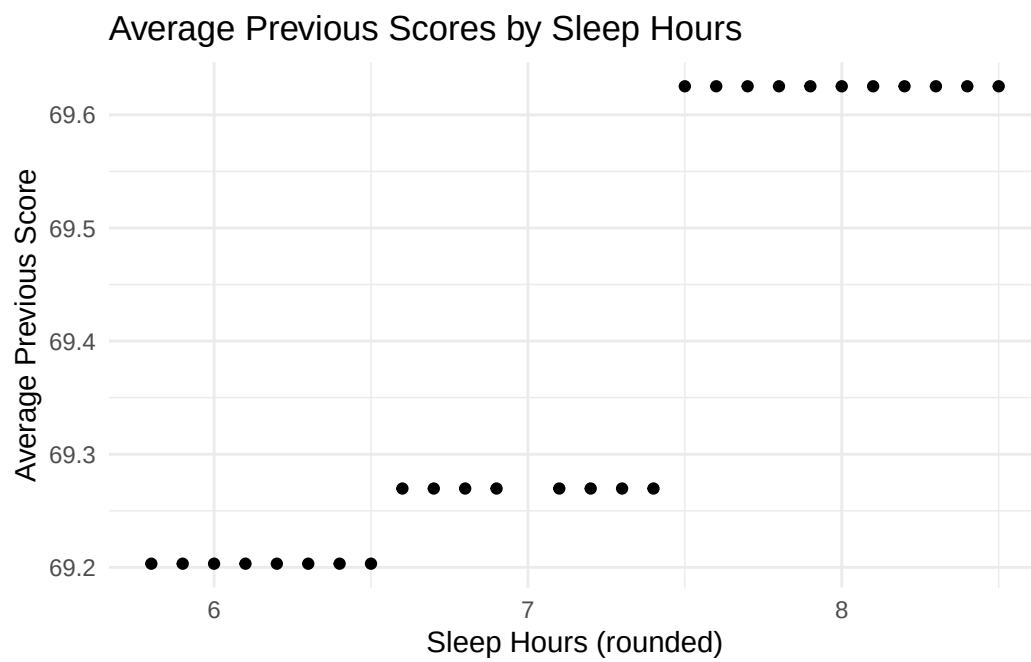


Research question 3

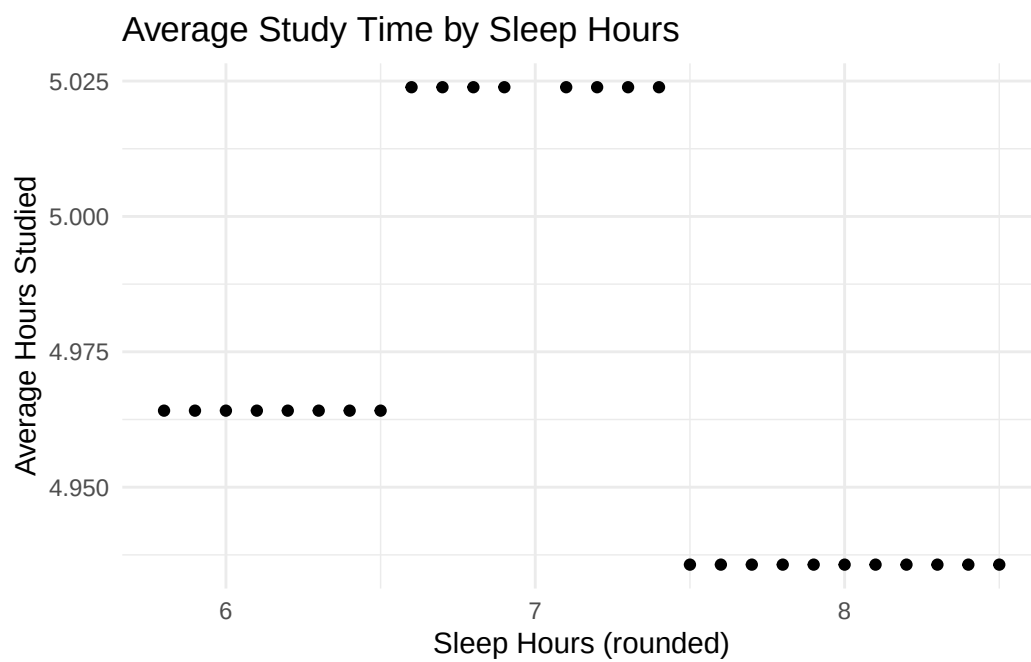
```
# Sleep hours vs academic performance index
ggplot(full_data, aes(x = sleep_duration, y = avg_performance)) +
  geom_point() +
  theme_minimal() +
  labs(
    title = "Average Performance Index by Sleep Hours",
    x = "Sleep Hours (rounded)",
    y = "Average Performance Index"
  )
```



```
# Sleep hours vs previous scores
ggplot(full_data, aes(x = sleep_duration, y = avg_previous_score)) +
  geom_point() +
  theme_minimal() +
  labs(
    title = "Average Previous Scores by Sleep Hours",
    x = "Sleep Hours (rounded)",
    y = "Average Previous Score"
  )
```

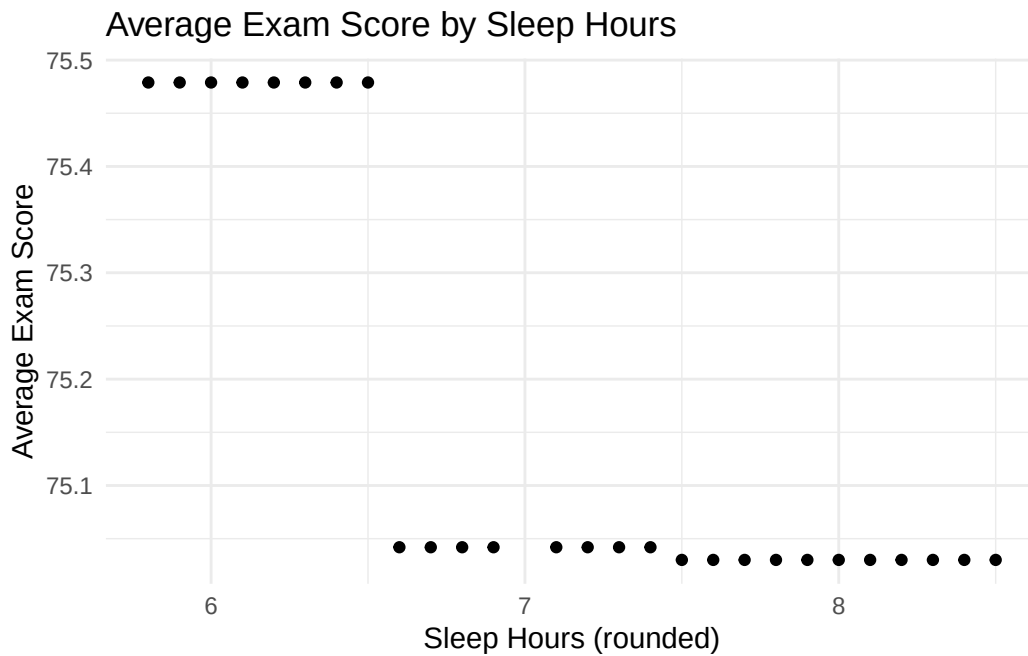



```
# Sleep hours vs hours studied
ggplot(full_data, aes(x = sleep_duration, y = avg_hours_studied)) +
  geom_point() +
  theme_minimal() +
  labs(
    title = "Average Study Time by Sleep Hours",
    x = "Sleep Hours (rounded)",
    y = "Average Hours Studied"
  )
```



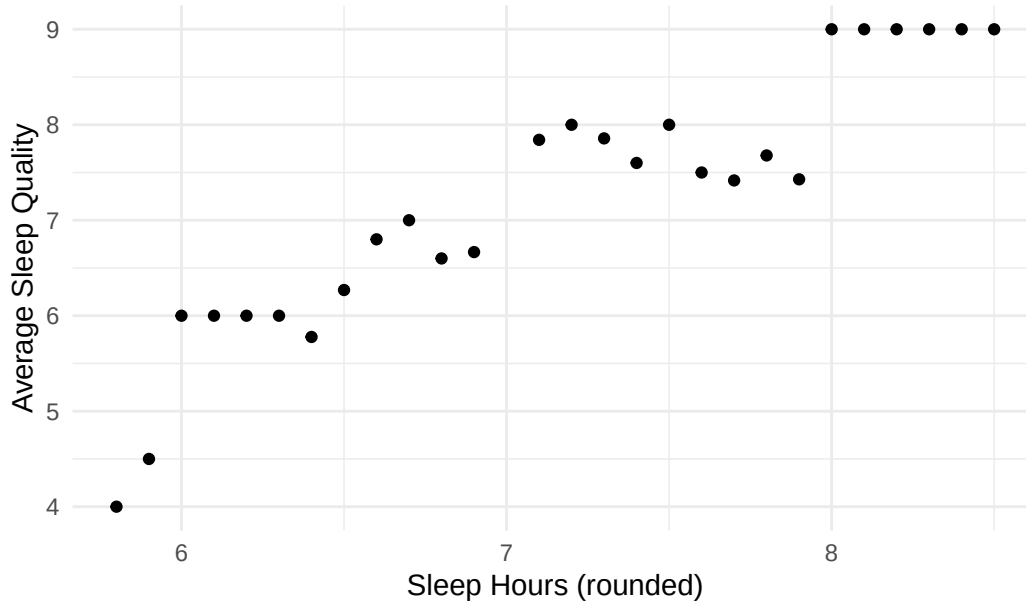
```
# Sleep hours vs exam score (from factors dataset)
ggplot(full_data, aes(x = sleep_duration, y = avg_score)) +
```

```
geom_point() +
theme_minimal() +
labs(
  title = "Average Exam Score by Sleep Hours",
  x = "Sleep Hours (rounded)",
  y = "Average Exam Score"
)
```



```
# Sleep hours vs sleep quality
ggplot(full_data, aes(x = sleep_duration, y = avg_sleep_quality)) +
  geom_point() +
  theme_minimal() +
  labs(
    title = "Average Sleep Quality by Sleep Hours",
    x = "Sleep Hours (rounded)",
    y = "Average Sleep Quality"
  )
```

Average Sleep Quality by Sleep Hours



```
# Sleep hours vs stress level
ggplot(full_data, aes(x = sleep_duration, y = avg_stress_level)) +
  geom_point() +
  theme_minimal() +
  labs(
    title = "Average Stress Level by Sleep Hours",
    x = "Sleep Hours (rounded)",
    y = "Average Stress Level"
  )
```

Average Stress Level by Sleep Hours

